SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2: GAME-BASED LEARNING (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO3	**Assessment:**	Assessment 2: Game-Based Learning (5 Questions × 10 Marks)
Weightage:	35% of CIA	**Total Marks:**	
CO1 Statement:	***learning algorithms using perceptro perceptrons, forward propagation, b gradient descent optimization techni Maximum LikelihoodEstimation for real-world problems.***		***mental concepts of machine learning and neural networks to design and analyze***
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Section / Batch:	AIML	**Date:**	

Code of Conduct

I D.Anand paul(192425207) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

QUESTIONS: 5 Total Marks: 50

Annexure A

Game-Based Learning

Game Title: CNN Quest – The Image Intelligence Challenge

Game Scenario:

Students act as AI Engineers responsible for building a CNN-based image classification system. They must complete different levels by correctly applying CNN concepts. Each level tests a different part of the CO3 syllabus.

Level 1 – CNN Fundamentals

1. What is the primary purpose of a CNN in image processing, and how is it different from a traditional Artificial Neural Network (ANN)?
2. Arrange the following CNN components in the correct order:
   Convolution Layer → Pooling Layer → Fully Connected Layer → Input Image → Output/Classification.
3. Identify the CNN component:
   "I scan across an image and detect patterns such as edges and textures. I use the same set of weights at different locations."
   Who am I?

Level 2 – Convolution & Filters

4. A 5 × 5 image is processed using a 3 × 3 kernel with a stride of 1 and no padding. What will be the size of the resulting feature map? Explain how you obtained the answer. 5. A CNN uses different kernels to detect edges, curves, textures, and other patterns. Explain why a CNN requires multiple filters instead of using only one filter throughout the network.
6. Match the following:

Component	Function
Kernel	?
Feature Map	?
Stride	?

Padding ?

7. Students must correctly match each component with its function.

Level 3 – Feature Map & Pooling Challenge

7. A CNN produces a feature map containing strong activation values around the edges of an object.
   What does this feature map tell us about the image, and why are feature maps important in CNNs?
8. A 4 × 4 feature map is given below:
9. 1 3 2 4
10. 5 6 1 2
11. 7 2 8 3
    4 5 6 9

Apply 2 × 2 Max Pooling with a stride of 2 and calculate the resulting feature

map. 12. Max Pooling vs Average Pooling:
Students are given different scenarios and must choose which pooling method would be more appropriate and justify their choice.

Level 4 – CNN Architecture Challenge

10. A CNN model performs extremely well on training images but gives poor results on new images.

Identify the problem and select the most appropriate solution from the following:

• Increase overfitting
• Remove all training data
• Use regularization/data augmentation
• Increase the image noise

Explain your choice.

11. Parameter Sharing Puzzle:
A convolution filter contains 9 weights and is applied across different regions of an image.

Explain how the same 9 weights can be reused at different locations and why this significantly reduces the number of parameters in a CNN.

Level 5 – Architecture Battle: AlexNet vs ResNet

12. Students are shown two simplified CNN architectures and must identify which represents AlexNet and which represents ResNet.
13. AlexNet vs ResNet Challenge:
Match the characteristics with the correct architecture:

• Introduced residual/skip connections
• Famous for ImageNet performance
• Helps solve vanishing-gradient problems in deep networks
• Uses a relatively deep CNN architecture for its time

14. Scenario Challenge:
You are given a complex image-classification problem requiring a very deep CNN.

Would you choose AlexNet or ResNet? Why?
Students must provide a justification based on architecture and training considerations.

Level 6 – CNN Real-World Mission

15. Students are given different applications:

• Face Recognition
• Self-Driving Cars
• Medical Image Diagnosis
• Handwritten Digit Recognition
• Object Detection
• Agricultural Disease Detection

Match each application with the image-processing task performed by CNNs.

16. Final Boss Challenge :
Design a basic CNN architecture for a cat-vs-dog image classification system. Students must specify:

• Input image
• Number/type of convolution layers

- Filters
- Pooling layers
- Fully connected layer
- Output layer
- Classification result

Level 1 – CNN Fundamentals

1. Purpose of a CNN vs. Traditional ANN

- **Primary Purpose:** The primary purpose of a Convolutional Neural Network (CNN) is to automatically learn and extract spatial features (like edges, shapes, and textures) directly from 2D image data without requiring human engineers to manually define those features.
- **Difference from ANN:** A traditional Artificial Neural Network (ANN) requires an image to be "flattened" into a 1D line of pixels immediately, which destroys the spatial relationships between neighboring pixels. A CNN preserves the 2D grid structure of the image and uses parameter sharing (filters), making it highly efficient for image processing.

2. Correct Order of Components

Input Image → Convolution Layer → Pooling Layer → Fully Connected Layer → Output/Classification.

3. Identify the Component

Answer: Filter (or Kernel).

Justification: A filter is a small matrix of weights that slides (scans) across the image to detect specific patterns. It reuses the exact same weights at every location to find that specific feature anywhere in the image.

Level 2 – Convolution & Filters

4. Feature Map Size Calculation

- **Answer:** The resulting feature map will be **3 × 3**.
- **Explanation:** The size of an output feature map is calculated using the formula: (W - K + 2P) / S + 1

Where **W** is the input width (5), **K** is the kernel size (3), **P** is padding (0), and **S** is the stride (1).

Calculation: (5 - 3 + 0) / 1 + 1 = 2 / 1 + 1 = **3**.

5. Why Multiple Filters are Required

A single filter can only detect one specific type of pattern (e.g., only horizontal lines or only red hues). Real-world images contain a complex combination of features. A CNN requires multiple filters in a single layer so it can simultaneously detect a wide variety of patterns—such as vertical edges, curves, and textures—building a complete visual understanding of the image.

6. Match Component to Function

• **Kernel:** A small matrix of weights that slides over the image to extract specific features. • **Feature Map:** The output matrix generated after applying a kernel, highlighting where specific patterns were detected.

• **Stride:** The number of pixels by which the kernel shifts across the image at each step. • **Padding:** Adding extra rows and columns (usually zeros) around the input image to preserve its spatial dimensions after convolution.

Level 3 – Feature Map & Pooling Challenge

7. Feature Map Interpretation

• **What it tells us:** A feature map with strong activations around object edges tells us that the specific kernel used was an "edge detector." It successfully located the boundaries separating the object from its background.

• **Importance:** Feature maps represent the network's internal understanding of the image. They translate raw pixels into abstract, meaningful concepts that the network uses to make its final decision. **8. Pooling Calculation**

Given the 4 × 4 feature map, applying **2 × 2 Max Pooling with a stride of 2** divides the matrix into four distinct 2 × 2 quadrants and extracts the highest value from each:

• Top-Left (1, 3, 5, 6) → Max is **6**

• Top-Right (2, 4, 1, 2) → Max is **4**

• Bottom-Left (7, 2, 4, 5) → Max is **7**

• Bottom-Right (8, 3, 6, 9) → Max is **9**

Resulting 2 × 2 Feature Map:

6	**4**
7	9

9. Max Pooling vs. Average Pooling

• **Max Pooling:** Selects the highest pixel value in a region. It is most appropriate when you want to highlight the most prominent, high-intensity features in an image (like sharp edges or bright anomalies). • **Average Pooling:** Calculates the mean of the pixels. It is more appropriate when you want to retain the overall average context or smooth out an image. *Max pooling is generally preferred for classification tasks.*

Level 4 – CNN Architecture Challenge

10. Troubleshooting Model Performance

- **Identify the Problem:** The model is suffering from **Overfitting**. It has memorized the training data but failed to learn general patterns for new images.
- **Appropriate Solution: Use regularization/data augmentation.**
- **Explanation:** Data augmentation alters the training data (e.g., flipping, rotating, or zooming) so the model cannot memorize exact pixel layouts. Regularization forces the model to learn robust, generalizable features that apply to unseen data.

11. Parameter Sharing Puzzle

In a CNN, the same 9 weights (a 3×3 kernel) are slid across the entire image. Instead of assigning a unique mathematical weight to every single pixel connection, the CNN assumes that a visual feature (like a curve) is just as important in the top-left corner as it is in the bottom-right. Reusing these 9 weights globally drastically reduces the total number of parameters, saving memory and computational power.

Level 5 – Architecture Battle: AlexNet vs ResNet

12. Identifying Architectures

- **AlexNet:** Visually represented as a standard, straight-line sequential architecture (Conv → Pool → Conv → Pool → FC).
- **ResNet:** Visually distinguished by its "skip connections" or "shortcut connections"—curved arrows bypassing layers to add the input back into the output deeper in the network.

13. AlexNet vs. ResNet Challenge

- **Introduced residual/skip connections:** ResNet
- **Famous for ImageNet performance:** AlexNet (Famous for triggering the 2012 deep learning boom) & ResNet (Famous for achieving human-level accuracy).
- **Helps solve vanishing-gradient problems in deep networks:** ResNet
- **Uses a relatively deep CNN architecture for its time:** AlexNet (8 layers was considered deep at the time).

14. Scenario Challenge: Very Deep CNN

- **Choice:** ResNet.
- **Justification:** When building a very deep CNN (e.g., 50+ layers), standard architectures like AlexNet suffer from the **vanishing gradient problem**, where the mathematical signals used to update weights become too small, causing early layers to stop learning. ResNet's skip connections act as "express lanes," allowing gradients to bypass layers and flow smoothly, ensuring stable training for deep

networks.

Level 6 – CNN Real-World Mission

15. Match Application with Image-Processing Task

• **Face Recognition:** Biometric identification and spatial facial feature matching. • **Self-Driving Cars:** Real-time semantic segmentation and object tracking (identifying roads, pedestrians, and signs).

• **Medical Image Diagnosis:** Anomaly classification (e.g., identifying tumors or pneumonia). • **Handwritten Digit Recognition:** Multi-class character classification (Optical Character Recognition / OCR).

• **Object Detection:** Bounding box localization and classification (finding exactly where an object is). • **Agricultural Disease Detection:** Leaf/crop texture and pattern classification.

16. Final Boss Challenge: Cat vs. Dog CNN Architecture

• **Input Image:** A 2D color image (e.g., 64 × 64 pixels with 3 RGB color channels). • **Convolution Layers:** 2 to 3 layers using 3 × 3 filters with ReLU activation to detect edges (whiskers, ears) and textures (fur).

• **Filters:** E.g., 32 filters in Conv Layer 1, and 64 filters in Conv Layer 2 to detect increasingly complex animal features.

• **Pooling Layers:** 2 × 2 Max Pooling layers applied after each convolution layer to reduce image dimensions and highlight the most prominent features.

• **Fully Connected Layer:** The 2D maps are flattened into a 1D vector and fed into a dense layer (e.g., 128 neurons) to learn the correlations between whiskers, snouts, and ears.

• **Output Layer:** A single neuron using a Sigmoid activation function.

• **Classification Result:** The output generates a probability score between 0 and 1 (e.g., a score < 0.5 classifies as "Cat", > 0.5 classifies as "Dog").

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightfultechnical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limitedaccuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach

Use of Tools	20	Advancedtools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of toolsandbasic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized,and professional communication	Understandable butlacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethicalawareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge Course Coordinator Program Director

Signature: Signature: Signature:

Date: Date: Date: